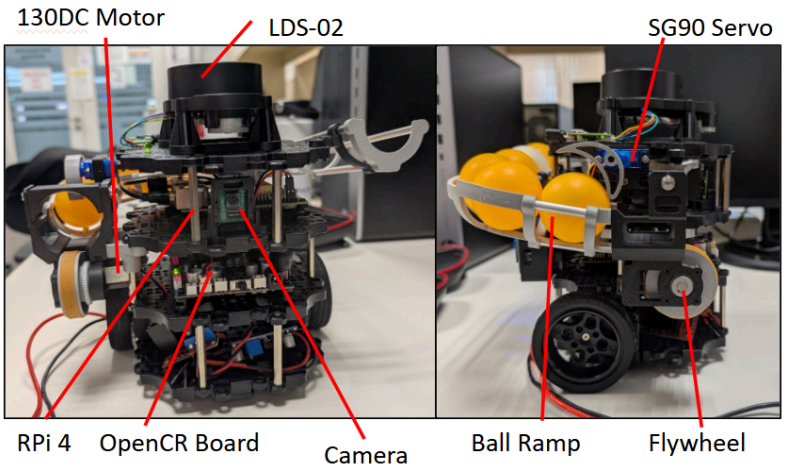
END USER DOCUMENTATION

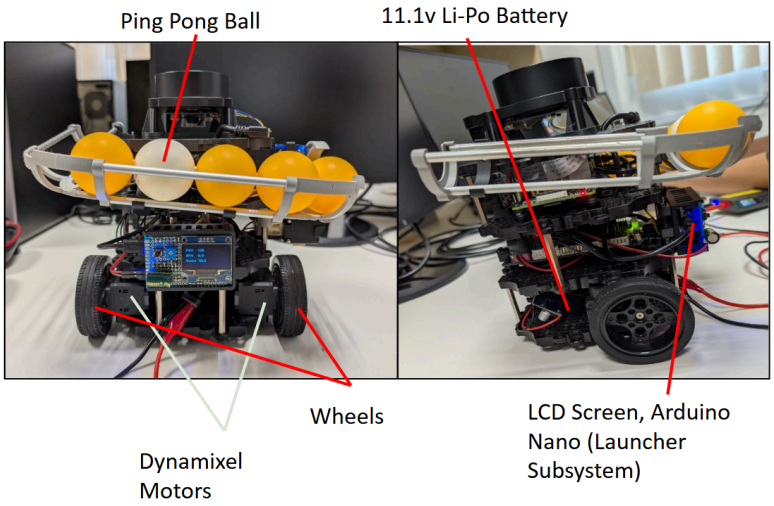
TurtleBot3 | Group 5 | CDE2310

Section 1 – General System Description & Critical Data (Spec Sheet)

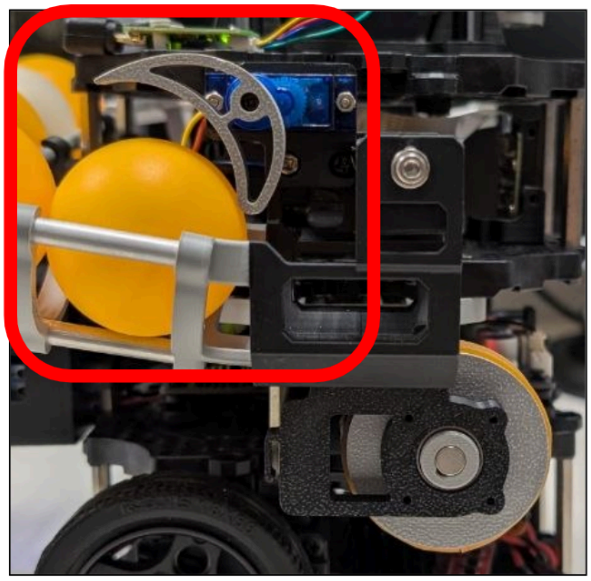
Model / Platform	TurtleBot3 Burger (Modified – G5)
Software Version	v1.1.0
ROS Version	ROS 2 Humble
Total Mass	~1095 g (including launcher assembly)
Centre of Gravity (x, y, z)	(−2.28 mm, −8.18 mm, 71.75 mm) from robot base centre
Battery	LI-PO 11.1V 1800mAH LB-12 19.98Wh 5C
Drive Motors	DYNAMIXEL XL430-W250 × 2
LiDAR	LDS-02 (360°, top-mounted)
Camera	Raspberry Pi Camera Module (forward-facing)
SBC	Raspberry Pi 4
MCU	OpenCR 1.0, Arduino Nano
Flywheel Motor	130 DC Motor 6V
Mission Time Limit	25 minutes (includes document check at start)
Operating Environment	Simulated Indoor Warehouse Conditions
Bot Purpose	Autonomous SLAM navigation + ping pong ball delivery to Station A (static ArUco) and Station B (moving ArUco)
Document Version	v1.1
Date of Issue	14 Apr 2026

Section 2 – Technical Guide





Servo gate idle position



2.1 How the TurtleBot3 Accomplishes the Mission

The robot runs ROS 2 Humble on a Raspberry Pi 4 (SBC), which communicates with the OpenCR 1.0 (MCU) over USB. The LDS-02 LiDAR feeds the Nav2 SLAM stack for autonomous mapping and navigation. The forward-facing camera detects ArUco markers to identify and align with delivery stations. The Raspberry Pi commands the launcher controller (Arduino Nano MCU) through USB to dispense ping pong balls on command.

2.2 Pre-Mission Start Procedure

#	Step
1	Charge and verify battery voltage ≥ 10.8 V. Power on robot via OpenCR switch.
2	Load 6 ping pong balls onto ramp.
3	Ensure servo gate encapsulates one ping pong ball.
4	Place the robot at the start position.
5	Begin screen recording of RViz.
6	On RPi (SSH) – run: **rosbu** (Wait till rosbu prints 'Run!' on the RPi)
7	On Remote PC – source workspace, then run: **ros2 run auto_nav mainlaunch** (on mission start command)
8	Confirm RViz is visible with active LiDAR scan.

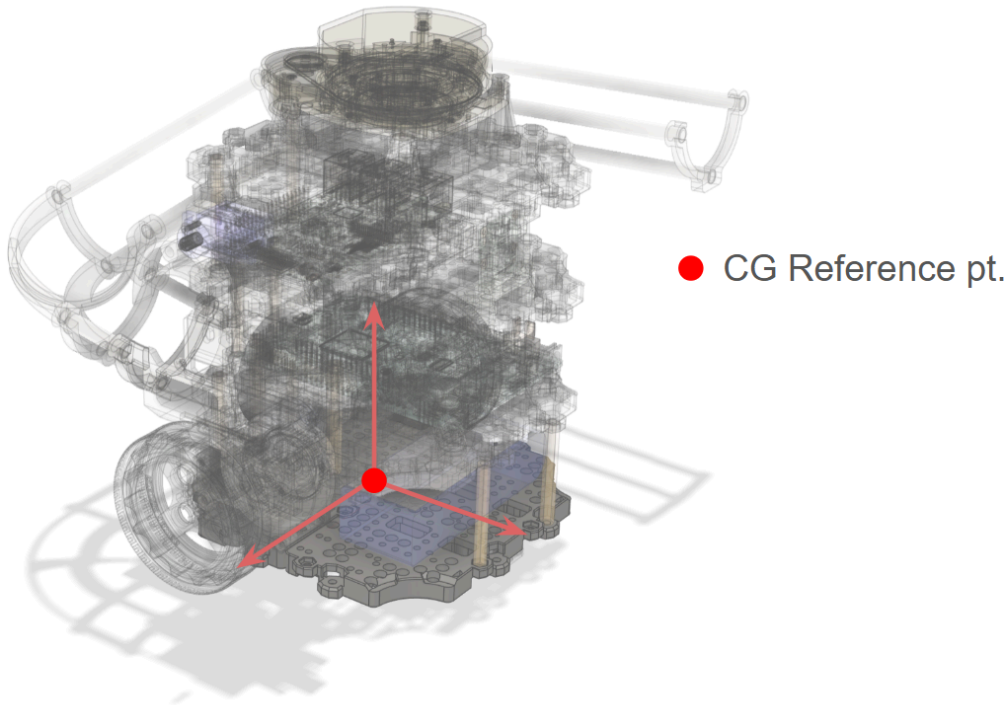
2.3 Mission Flow

Step	Phase	Description
1	Explore & Map	SLAM + frontier search. Target ≥ 80% map coverage. Uses ≤ 6 landmark markers.
2	Station A	Detect static ArUco. Align within 10 cm. Dispense exactly 3 balls in sequence.
3	Station B	Differentiate moving vs. static ArUco. Track position. Dispense onto moving target.
4	Recovery	Nav2 auto re-plans on failure with no human intervention.

2.4 Key Software Nodes

Node	Function
exploration2 node	Path planning, obstacle avoidance, recovery behaviours Online SLAM – map generation and localisation
aruco_detector2 node	Marker detection; static vs. dynamic classification
fsm_controller node	Top-level state machine; sequences all tasks autonomously
launcher node	Flywheel speed control and servo gate logic

2.5 Launcher Assembly Specs



Component	Detail
Flywheel Motor	130 DC Motor; driven viaL298N driver; position (−75, 5, 67) mm
Servo Gate	SG90 servo; releases one ball at a time; position (−75, −18, 130) mm
Ramp Guides	6 mm OD aluminium tubes × 6; length 250 mm each
Motor Driver	L298N Motor driver, powered by 9V Battery
Target Success Rate	≥ 70% ball dispense accuracy

Section 3 – Acceptable Defect Log

Record defects that are not in perfect condition but still acceptable for mission execution. Must be assessed and signed off by a team member before mission start.

#	Component	Description of Defect	Justification / Mitigation	Assessed By	Date
1	SG90 Servo	Slight servo jitter, < 5 degrees from nominal position	Servo jitter due to EMI from inrush current during startup.	3SG Wang yizhang	14 Apr 2026
2	Ping pong ball ramp	Guide rods may become disconnected from frame	Rod disconnection does not occur during normal operation; only when the robot is lifted by the ramp. Reinsertion takes <10 seconds and does not affect launcher accuracy	Niki S	14 Apr 2026

Assessed By: 3SG Gary Chen Date: 14 Apr 2026
Signature: _____

Section 4 – Factory Acceptance Test (FAT) Checklist

Complete all checks before the mission run. Part of the 25-minute mission window

Hardware

#	Check Item	P / F
H1	Battery voltage ≥ 10.8 V before power-on	P / F
H2	All structural plates securely fastened	P / F
H3	No loose cables dangling outside of robot chassis	P / F
H4	DYNAMIXEL motors and wheels rotate freely, with no binding, resistance or obstruction	P / F
H5	LiDAR spins freely with no jam or abnormal resistance	P / F
H6	Flywheel and servos launch ping pong balls with no jam or abnormalities Launch command on RPi: **rosbu** Launch command on Remote PC: **ros2 run auto_nav fat_launch**	P / F
H7	ArUco marker detected at 0.5m at maze lighting conditions Launch command on RPi: **rosbu** Launch command on Remote PC: **ros2 run auto_nav fat_aruco**	P / F
H8	Ping pong balls slides smoothly down ramp	P / F
H9	Servo gate encapsulates 1 ping pong ball in idle position	P / F

Software & Communication

#	Check Item	P / F
S1	Remote PC ↔ RPi SSH connection established	P / F
S2	LIDAR data received & plotted in active RViz map after launch	P / F
S3	No error logs or messages during testing of hardware	P / F
S4	Ctrl-C terminates all nodes	P / F

FAT Sign-off – Test Conductor: _____ Date: _____

Section 5 – Maintenance and Part Replacement Log

Record all maintenance and component replacements. This log stays with the robot at all times.

5.1 Maintenance Log

#	Date	Component	Action Taken / Part Replaced	Reason	Done By
1	14 Apr 2026	9V Battery Cell	9V Battery Added	Isolate Launcher Power Supply	3SG Gary Chen